CHHATRAPATI SHIVAJI MAHARAJ INSTITUTE OF TECHNOLOGY

Affiliated to University of Mumbai, B++ NAAC accreditation

CODE : AUTOMATA

VER. 2.1

TEAM DETAILS

- **Problem Statement:** Telemedicine Access Platform for Rural Healthcare – Develop a digital platform to improve healthcare access for rural populations.
- **PS No. :** HC-1
- **College Name :** Chhatrapati Shivaji Maharaj Institute of Technology, Panvel.
- **Team Name :** NaN-Chalant
- **Team Members :**
  1. Gaurangi Madankar
  2. Vaibhav Kadam
  3. Tanvi Gandhi
  4. Snehal Mohite

IDEA APPROACH

Phase A: Intake & Triage (The Digital Front Door)
 * User Entry: Patient calls the Helpline.
*IVR Logic:
* Press 1 (Emergency): Immediate bridge to Human Doctor (Priority).
 * Press 2 (General): Connects to AI Voice Agent.
 * Press 3 (Follow-up): Status check on existing orders.
AI Assessment:
 * Standard: AI collects symptoms, suggests home remedies, and generates a case summary.
 * Red Flag: If symptoms are suspicious/severe, AI triggers a "Priority Callback" request (Doctor calls back within 10 mins).

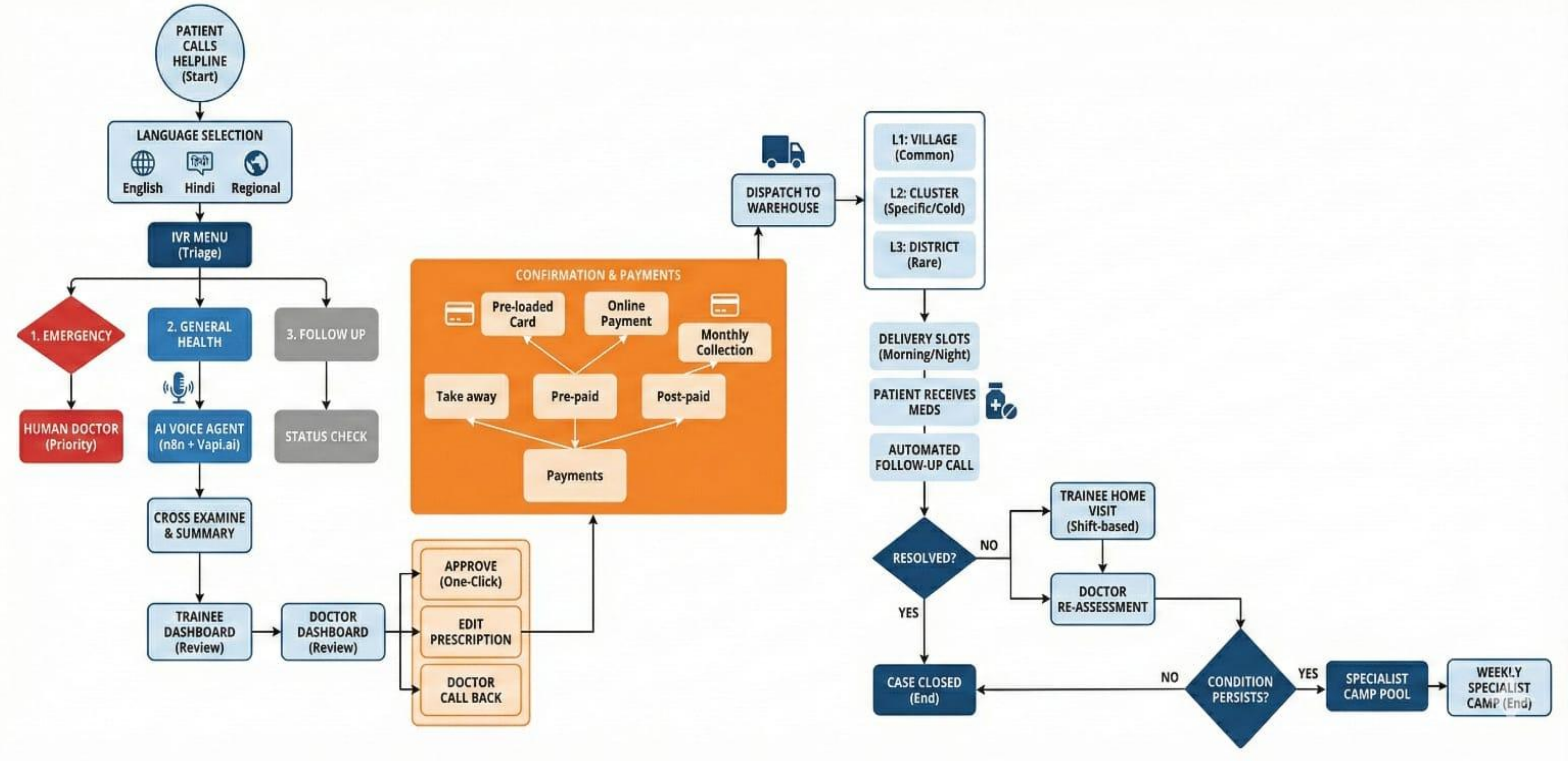
Phase B: Consultation & Logistics (The Execution)
 * Doctor Dashboard: Doctor reviews AI summary.
 * Action: Edits prescription or One-Click Approves.
Order Confirmation: Automated AI call to the patient.
* Confirms: "Medicines cost ₹150. Confirm?"
* Payment Mode: User selects Cash (Postman), UPI (WhatsApp), or Wallet Balance.
Inventory Logic: System checks stock hierarchically: Village Cluster District.
* Dispatch (India Post):
* Slotting: Orders < 12 PM = Night Slot. Orders > 12 PM = Next Morning Slot.
* Delivery: Postman (GDS) scans packet via RICT Device Collects Payment Handover.

Phase C : The Safety Net (Automated Loop)
 * Follow-Up Call: AI calls patient 3 days later.
 * "Feeling Better?": Yes Case Closed.
 * "No Change?": Ticket Escalated to Phase D.

Phase D: Medical Escalation (Physical Visit)
*Trainee Dispatch: System assigns a Medical Intern (Trainee) for a home visit (Morning/Night shift).
* Vitals Check: Trainee captures BP, Oximeter, and Physical Exam data, and uploads to Dashboard.
* Re-Assessment: Main Doctor reviews new vitals Prescribes stronger medication. Cycle repeats (Postman delivery).

Phase E: The Specialist Camp (Final Tier)
*Camp Pooling: Unresolved cases are added to the "Specialist Pool" database.
* Weekly Logic: Camp held at Cluster Center (every Sunday).
* Coordination: AI Agent calls pooled patients: "Dr. Sharma (Cardiologist) is coming this Sunday. Confirm appointment?

ARCHITECTURE/FLOWCHART



USE CASES / BUSINESS CANVAS

KEY PARTNERS

- Government Health Centers (PHC, CHC, Ayushman Centers)
- Licensed Doctors and Specialists
- Medical Interns / Trainees
- India Post or Local Delivery Workers
- Medicine Suppliers and Pharmaceutical Distributors
- Cloud Service Providers (Supabase)
- AI Voice API Providers (Vapi, Speech-to-Text providers)
- Government Health Departments (for approvals and scaling)

KEY ACTIVITIES

- AI-based patient triage
- Doctor case review & prescription approval
- Medicine inventory management (3-level hierarchy)
- Medicine dispatch & logistics coordination
- Trainee home visit scheduling
- Weekly specialist camp coordination
- Data tracking & analytics
- Follow-up automation

KEY RESOURCES

- AI Voice System
- Doctor Dashboard (Web App)
- Trainee Android App
- Central Database
- Medicine Inventory Stock
- Call Center Infrastructure
- Trained Medical Staff
- Cloud Servers

VALUE PROPOSITIONS

- One-call healthcare access for rural villagers
- AI-assisted triage system that reduces doctor workload
- 12-hour medicine delivery system
- Multi-stage escalation model (AI → Doctor → Trainee → Specialist Camp)
- Hierarchical inventory management to reduce cost and wastage
- Continuous follow-up mechanism to ensure case resolution
- Reduced need for patients to travel to urban hospitals
- Affordable, scalable, and structured rural healthcare solution

CUSTOMER RELATIONSHIP

- AI follow-up calls
- Human doctor supervision
- Trainee home visits
- Community health camps
- Local language communication
- Health education calls

CHANNELS

- Helpline number (IVR-based system)
- AI voice interaction system
- Medicine home delivery service
- Weekly specialist camps
- Trainee home visit services

CUSTOMER SEGMENTS

- Rural villagers
- Elderly and chronic patients
- Government rural health programs
- Low-income families
- Public health departments (for disease monitoring and planning purposes)

COST STRUCTURE

- AI API usage cost
- Cloud hosting cost
- Doctor consultation fees
- Trainee stipends
- Delivery logistics cost
- Medicine inventory cost
- Call center operational cost
- Maintenance & system upgrades

REVENUE STREAM

- Government funding and public health grants
- Small service fee per consultation (minimal and affordable)
- Margin on medicine distribution
- Corporate Social Responsibility (CSR) partnerships
- Health data analytics partnerships (aggregated and anonymized data only)
- Subscription model for district-level or state-level deployment

TECHNOLOGY USED

Core Infrastructure

Backend: FastAPI (Python)

Frontend: React (Vite) + Tailwind CSS

Orchestration: n8n (Local/Self-hosted)

AI & Communication

Voice Engine: Vapi.ai (Real-time AI Voice)

AI Brain: Google Gemini 1.5 Flash (RAG & Diagnostics)

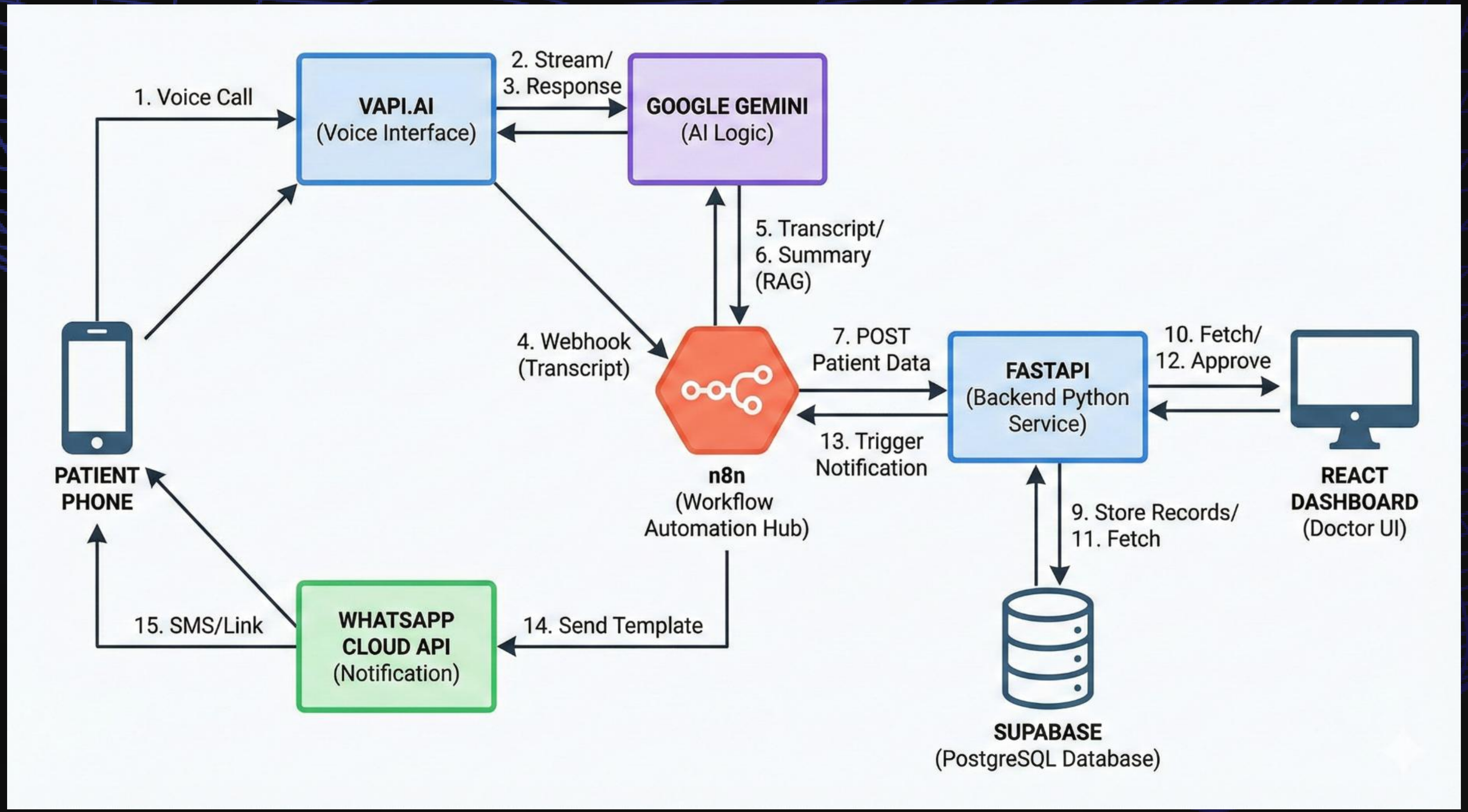
Messaging: WhatsApp Cloud API (Meta))

Specialized Features

Vector Search: Supabase pgvector (for Medical Dataset RAG)

Database: Supabase (PostgreSQL)

Hosting: Vercel (Frontend) & Local Tunnel/ngrok (for n8n/FastAPI demo)



- RAG (Retrieval-Augmented Generation): that the AI isn't just "guessing"—it’s looking up real Kaggle medical data.
- Asynchronous Processing: FastAPI handles multiple requests simultaneously without slowing down.
- Vector Embeddings: Supabase (pgvector) is used to store and search medical symptoms.Regional Language
- Processing: that Gemini handles the translation from English to Hindi/Regional languages on the fly.